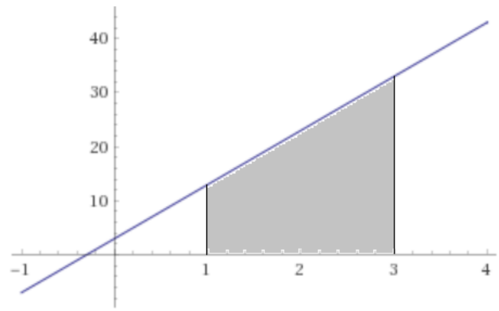
## 5.2.3 Average Value Worksheet

1. Determine the average value of the function $f(x)=10x+3$ on the interval $[1,3]$.

$$f_{avg}=\frac{1}{3-1}\cdot\int_1^3(10x+3)\,dx=\frac{1}{2}\cdot\int_1^3(10x+3)\,dx$$

We can use geometry to find the area under $f(x)=10x+3$ on the interval $[1,3]$.

$$A=\int_1^3(10x+3)\,dx=2(13)+\frac{1}{2}(2)(20)=26+20=46$$



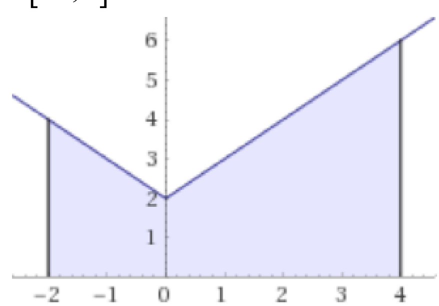

Therefore we have $f_{avg}=\frac{1}{2}\cdot\int_1^3(10+3)\,dx=\frac{1}{2}(46)=23$ .

2. Determine the average value of the function $f(x)=|x|+2$ on the interval $[-2,4]$.

$$f_{avg}=\frac{1}{4-(-2)}=\int_{-2}^4(|x|+2)\,dx=\frac{1}{6}\cdot\int_{-2}^4(|x|+2)\,dx$$

We can use geometry to find the area under $f(x)=|x|+2$ on the interval $[-2,4]$.

$$A=\int_{-2}^4(|x|+2)\,dx=6(2)+\frac{1}{2}(2)(2)+\frac{1}{2}(4)(4)=12+2+8=22$$




Therefore we have $f_{avg}=\frac{1}{6}\cdot\int_{-2}^4(|x|+2)\,dx=\frac{1}{6}(22)=\frac{11}{3}$.

3. Find all numbers $c$ that satisfy the MVT for Integrals for the function $f(x)=10x+3$ on the interval $[1,3]$.

Since $f(x)=10x+3$ is continuous on $[1,3]$, the MVT for integrals states that there is a $c$ in $[1,3]$ such that:

$$f(c)=\frac{1}{2}\cdot\int_1^3(10x+3)\,dx=23$$

So we have $10c+3=23 \Rightarrow 10c=20 \Rightarrow c=2$.

4. Find all numbers *c* that satisfy the MVT for Integrals for the function $f(x)=|x|+2$ on the interval $[-2,4]$.

   Since $f(x)=|x|+2$ is continuous on $[-2,4]$, the MVT for integrals states that there is a *c* in $[-2,4]$ such that:

$$f(c)=\frac{1}{6}\cdot\int_{-2}^{4}(|x|+2)\,dx=\frac{11}{3}$$

   So we have $|c|+2=\frac{11}{3} \Rightarrow |c|=\frac{5}{3} \Rightarrow c=\pm\frac{5}{3}$.

   Since both values are in the interval $[-2,4]$ they are both acceptable values of *c*.

5. Suppose the function $k(x)$ is continuous on the interval $[1,8]$ and $\int_{1}^{8}k(x)\,dx=28$. Show that the function $k(x)$ assumes the value of 4 at least once on the given interval.

   By the Mean Value Theorem we know that there must exist a number *c* in $[1,8]$ such that

$$k(c)=\frac{1}{8-1}\cdot\int_{1}^{8}k(x)\,dx=\frac{1}{7}(28)=4$$

6. Suppose that a car with whose velocity (in feet per second) after *t* seconds is given by $v(t)=3t^2+4t-1$. Show that the car must have been moving at a speed of 135 *ft/sec* at some point during the interval $[4,8]$

   We can see that since $v(t)$ is a polynomial is must be continuous and differentiable on $[4,8]$. Therefore, by the Mean Value Theorem for Integrals we know that there must exist a number *c* in $[4,8]$ such that

$$v(c) = \frac{1}{8-4}\int_4^8 v(t)\,dt = \frac{1}{4}\left(x^3 + 2x^2 - x\right)\Big|_4^8 = \frac{1}{4}\left[632 - 92\right] = \frac{1}{4}\left[540\right] = 135$$